

Rolling Airline Recovery under Evolving Information: An Interface-Consistent Decision Chain

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Abstract

1 Introduction

Airline disruption recovery is a rolling decision problem in which recovery choices must be repeatedly updated as operational information evolves. Unlike static optimization problems, recovery decisions are made under interconnected operational processes involving aircraft continuity, turnaround operations, airport conditions, passenger exposure, and downstream execution opportunities [Hassan et al., 2021, Evler et al., 2022]. Therefore, the same observed delay may correspond to different recovery requirements, and a recovery decision cannot be formed from the latest delay observation alone. It depends on how evolving operational information is transformed into a decision-relevant state and how the dependencies represented by that state are preserved throughout subsequent recovery decisions.

Existing research has established strong foundations for dynamic and uncertainty-aware airline recovery. Rolling-horizon methods update recovery plans as new disruption information becomes available [Vink et al., 2020]. Stochastic formulations represent uncertain disruption evolution and information revelation processes [Zhao et al., 2023]. Predictive information has also been incorporated into downstream airline planning and recovery decisions [Lee et al., 2020, Birolini and Jacquillat, 2023]. In addition, industrial decision-support systems integrate heterogeneous operational data with large-scale recovery optimization methods [Davies et al., 2026]. These studies have substantially improved recovery decisions by refining individual decision components, enhancing information quality, and expanding the scope of recovery optimization.

Recent research has advanced operational information sharing, integrated recovery, predictive-prescriptive planning, and state-dependent decision making [EUROCONTROL, 2017, ?, ?, Hassan et al., 2021, Lee et al., 2020, Bellman, 1957, Kaelbling et al., 1998]. These streams establish important capabilities for making operational information available, coordinating coupled recovery decisions, and linking evolving observations to actions. Nevertheless, they typically begin from an already defined operational or decision state, or treat evidence qualification, state construction, consequence evaluation, and action comparison as separate modeling concerns. Thus, the cross-stage interfaces that determine what information is admissible at decision time, which dependencies must be retained, and whether actions are compared on a common consequence basis remain less explicitly formalized [Zhao et al., 2023, Ogunsina et al., 2022, Davies et al., 2026].

This issue is particularly important because airline recovery is characterized by sequential and dependent operational processes. A disruption affecting an upstream flight may alter downstream departure conditions, aircraft availability, turnaround opportunities, and passenger consequences. Consequently, the operating state used for recovery decisions is not a direct observation but a representation constructed from evolving information. If different stages independently reconstruct their inputs according

to local objectives, decision-relevant information may be omitted and previously established state dependencies may be weakened. The resulting recovery evaluations may then correspond to different operational realizations rather than alternative decisions for the same recovery situation.

This study addresses two questions. First, which decision-relevant information and state dependencies should be retained when evolving operational information is converted into a recovery state? Second, how should the retained information, the resulting state, and the available recovery actions be combined to generate a consistent recovery decision? We represent this process through two linked mappings:

$$\mathcal{E}_t \longrightarrow \mathcal{S}_t, \quad (\mathcal{E}_t, \mathcal{S}_t, \mathcal{A}_t) \longrightarrow \mathcal{D}_t,$$

where $\mathcal{E}_t$ denotes the information available at decision time t , $\mathcal{S}_t$ the operating state formed from this information, $\mathcal{A}_t$ the available recovery action set, and $\mathcal{D}_t$ the resulting recovery comparison. The first mapping describes how evolving information forms a decision-relevant state, while the second describes how the state, remaining decision-relevant information, and available actions jointly determine recovery decisions.

The contributions of this paper are threefold:

- We formulate a cross-stage decision chain for rolling airline recovery that explicitly represents how operational information is transformed into recovery states, how decision-relevant information and derived objects are transmitted and aligned across stages, and how states, information, and available actions jointly determine recovery decisions. Through a set of mapping functions, the same decision chain can be applied under different monetary systems and decision-authority scopes.
- We introduce an interface-consistency perspective for airline recovery. The proposed formulation specifies requirements for decision-time information availability, state-dependence preservation, common consequence representation, explicit support tracking, and decision authority across recovery stages.
- We operationalize the proposed decision chain using an airline recovery methodology and evaluate how alternative information-sharing and state-dependence designs affect state construction, consequence evaluation, action selection, and the managerial trade-offs among operational continuity, service consequences, resource use, and residual risk.

The remainder of the paper reviews the related literature, formalizes the proposed decision chain, describes its empirical implementation, and evaluates its methodological and operational performance.

2 Related Literature

2.1 Operational Information Sharing and the Shared Operating Picture

Information sharing is a core operational principle in collaborative aviation decision making. Airport Collaborative Decision Making (A-CDM) requires airport operators, aircraft operators, ground handlers, air navigation service providers, and the Network Manager to exchange accurate and timely information so that decisions are based on a more consistent operating picture [EUROCONTROL, 2025, Okwir et al., 2017]. Successive turnaround milestones also link inbound and outbound operations. As a milestone changes, downstream estimates, alerts, and notifications can be revised accordingly [EUROCONTROL, 2025]. Information sharing therefore affects both what operators know and when downstream expectations are updated.

Airline disruption recovery extends this coordination problem beyond airport milestones. Recovery decisions couple aircraft rotations, passengers, crew, maintenance, airport capacity, and other operational resources [Clausen et al., 2010, Hassan et al., 2021, Su et al., 2021, Wu et al., 2025]. Recent work also coordinates irregular-flight recovery using information exchanged between airlines and the

Network Manager [Wu et al., 2026]. Information sharing is thus well established in both operational practice and disruption-recovery research.

In the remainder of this paper, *cross-stage information sharing* refers to the retention and reuse of decision-relevant information across computational stages of one recovery process, rather than to information exchange between operational stakeholders. This distinction separates the methodological question studied here from the inter-organizational coordination problem addressed by A-CDM.

The unresolved issue is narrower than information availability. A shared operating picture does not specify how accumulated information should retain its decision role after it enters a recovery model. Historical observations may be needed to form the current recovery state, whereas current timing, identity, resource, or execution information may remain directly relevant to action formation and comparison. A rolling recovery methodology must therefore organize these different information roles across successive decision representations.

2.2 Distributional Representation and Decision-Relevant Information

Probabilistic models can preserve decision-relevant uncertainty that a point estimate would suppress. Distributional flight-delay models represent conditional uncertainty in arrival or departure outcomes rather than reducing an unresolved operation to a single expected value [Zoutendijk and Mitici, 2021, Dalmau et al., 2024]. This distinction matters when recovery consequences depend on threshold crossings, such as exhausted turnaround buffers, missed connection margins, or closed execution windows. Two operating states with similar expected delay can therefore create different recovery exposure.

The value of a probabilistic representation extends beyond forecast accuracy. Probabilistic delay predictions have been passed directly to flight-to-gate assignment, while probabilistic estimates of future airport disruptions have been incorporated into dynamic stochastic airline recovery [Zoutendijk and Mitici, 2021, Lee et al., 2020]. Predictive-prescriptive studies similarly embed data-driven delay or uncertainty estimates in aircraft-routing and scheduling decisions [Birolini and Jacquillat, 2023, Pang et al., 2024]. These studies establish that probabilistic output can remain useful after prediction rather than being collapsed before the decision stage.

Preserving uncertainty is not the same as preserving state dependence. Separate marginal distributions can retain uncertainty for individual quantities while discarding dependence among them. Bayesian delay-propagation models explicitly represent dependencies among operational variables, and relational dynamic Bayesian networks propagate uncertainty across disruption-management states [Wu and Law, 2019, Ogunsina et al., 2022]. For the decision-chain problem considered here, when several unresolved quantities jointly define an operating state, their joint or scenario-level structure must remain identifiable if downstream consequence and action calculations are to refer to the same operating realization. Distributional representation can therefore support cross-stage information sharing, but it does not by itself guarantee cross-stage state consistency.

2.3 State Dependence in Dynamic and Integrated Recovery

Dynamic and stochastic recovery has established that airline recovery decisions depend on an evolving operating state. Rolling-horizon methods recompute aircraft-recovery solutions as new disruptions occur [Vink et al., 2020]. Two-stage and scenario-based formulations account for uncertain disruption duration and the timing of improved disruption information [Zhao et al., 2023]. Dynamic stochastic recovery also combines realized disruptions with probabilistic information about future airport conditions [Lee et al., 2020]. These studies provide a strong basis for repeated decisions under progressively revealed information.

Operational dependencies make this state dependence consequential. Disturbances propagate through aircraft rotations and connected airport processes, while turnaround-aware recovery shows that the consequence of a delay depends on local flexibility and downstream conditions [Fleurquin et al., 2013,

Evler et al., 2022]. Integrated recovery further couples aircraft, crew, passenger, and schedule decisions. The feasibility and value of an action therefore depend on the recovery instance in which it is applied [Arikan et al., 2016, Maher, 2016, Ding et al., 2023]. Reinforcement-learning formulations express the same idea by conditioning recovery actions on a current disruption state [Ding et al., 2023].

The gap is not a lack of state dependence in prior models. Existing formulations preserve the dependencies required within their own forecasting, propagation, stochastic-optimization, or state-action structures. The less explicit issue arises when separate modeling functions contribute sequentially to one rolling decision. A history-conditioned operating realization may be represented probabilistically, translated into consequences, and then modified by candidate recovery actions. Cross-stage consistency therefore requires that these transformations continue to refer to the same underlying operating realization.

2.4 From Information Integration to Decision-Chain Consistency

Integrated recovery and industrial decision support demonstrate that heterogeneous information, resources, and recovery actions can be combined at operational scale. Airline-recovery formulations already cover retiming, cancellation, aircraft and crew rerouting, passenger reaccommodation, departure holding, and cruise-speed control [Arikan et al., 2016, Maher, 2016, Ding et al., 2023]. The Lufthansa Group decision-support system jointly optimizes aircraft, passenger, and crew recovery while drawing on operational schedule and maintenance information from multiple airline systems [Davies et al., 2026]. Collaborative irregular-flight recovery has also been extended to coordinated airline-Network Manager decisions [Wu et al., 2026].

These advances shift the methodological question from integration to interface consistency. Information availability does not determine which information should be retained through a history-conditioned state and which should remain directly reusable downstream. Probabilistic prediction can preserve uncertainty, but it does not ensure that later consequence models refer to the same operating realization. Integrated recovery can compare broad action sets, but a broad action domain does not by itself determine whether the downstream comparisons use the same consequence scope or whether the evidential status of action effects remains appropriately bounded.

The remaining gap is therefore a decision-chain consistency problem. In the closest recovery literature, the individual capabilities needed for dynamic updating, uncertainty-aware state representation, integrated action evaluation, and operational data integration are already well established. What is less explicit is the common interface governing how one stage feeds the next. For a rolling recovery comparison to remain internally consistent, information should enter only after it becomes decision-time admissible; a coherent unresolved operating realization should remain identifiable as it is transformed downstream; competing actions should be evaluated on a common consequence scope; and support or provenance should not be silently strengthened by numerical transformation. Section 3 formalizes these requirements. The resulting separation between the recovery state and the method-level action library also permits new action specifications to use the same interface, but this extensibility is a property of the decision-chain design rather than the research gap itself.

3 Cross-Stage Information Sharing and State-Dependent Recovery

Section 2 identifies a decision-chain consistency problem in rolling airline recovery. The issue is not whether operational information is available, but whether the information needed for recovery remains usable when the calculation moves from state formation to consequence evaluation and action comparison. Historical operational information is needed to characterize the current recovery situation, while current timing, resource, execution, and support information may remain directly relevant when recovery actions are formed and compared.

State dependence imposes a second requirement. Once the predecessor-successor operation has been

represented at a decision time, the downstream consequence calculation and the comparison of recovery actions should continue to refer to that same operating situation. The methodology below therefore keeps the information boundary, operating scenario, consequence basis, and action-comparison basis explicit. Sections 3.1–3.4 define the recovery model, and Section 3.5 summarizes the decision-chain consistency conditions implied by these definitions.

3.1 Cross-Stage Information Use and Decision-Chain Structure

Consider a predecessor–successor recovery episode $i = (f_i^-, f_i^+)$ for the same aircraft, observed at rolling decision time t . The predecessor affects when the aircraft becomes available for the successor; the successor then faces a remaining turnaround window, off-block execution, taxi conditions, downstream schedule exposure, passenger consequences, and a set of recovery opportunities. Following the notation fixed in Section 1, the decision chain is

$$\mathcal{E}_t \longrightarrow \mathcal{S}_t, \quad (\mathcal{E}_t, \mathcal{S}_t, \mathcal{A}_t) \longrightarrow \mathcal{D}_t. \quad (1)$$

For an individual episode, the subscript i is added to the corresponding objects. $\mathcal{E}_{i,t}$ contains the operational information available at time t ; $\mathcal{S}_{i,t}$ is the recovery state formed from the admissible operating history; $\mathcal{A}_{i,t}$ is the set of recovery actions available for the episode; and $\mathcal{D}_{i,t}$ is the resulting recovery comparison and recommendation output.

Table 1: Principal objects in the rolling airline-recovery decision chain.

Object	Airline-recovery meaning	Role in the decision chain
$\mathcal{E}_{i,t}$	Operational information available at decision time t	Provides the information boundary for state formation, action availability, and comparison eligibility
$\mathcal{H}_{i,t}$	Admissible operating history up to t	Retains predecessor progress, airport and weather evolution, and turnaround-related history used to form the current state
$\mathcal{S}_{i,t}$	History-conditioned unresolved operating state	Represents the remaining uncertainty of the predecessor–successor operation
$\mathbf{Q}_{i,t,s}^0$	Baseline operational-consequence profile for scenario s	Converts one operating scenario into flight, passenger, and recovery-resource consequences
$\mathcal{R}$	Recovery-action library	Defines the candidate recovery actions that may become available in an episode
$\mathcal{A}_{i,t}$	Recovery actions available at decision time t	Supplies the alternatives that can be evaluated for the current episode
$C_{i,t,s}^{a,\text{CU}}$	Action-conditioned consequence in constructed units	Places alternative actions on the same scenario-specific consequence basis
$\mathcal{D}_{i,t}$	Recovery comparison and recommendation output	Reports operating context, state/consequence exposure, comparable actions, ranking, and support status
Π_a	Declared support basis for action response a	Constrains how an action result can enter and be interpreted in formal comparison

Equation (1) does not require all operational information to be compressed into the state. Information describing recent operating evolution is retained through $\mathcal{H}_{i,t}$ and $\mathcal{S}_{i,t}$. By contrast, current flight identity, remaining execution time, resource availability, and action-specific support can remain directly available when $\mathcal{A}_{i,t}$ and $\mathcal{D}_{i,t}$ are formed. This distinction is important in airline recovery because the information needed to describe the current operating situation is not identical to the information needed to decide whether a particular intervention is still executable.

At each decision node, the model therefore provides three operational views rather than only a recommended action. First, it records the current operating context and information cutoff. Second, it reports the recovery state and the associated flight, passenger, and resource consequences. Third, it reports the available and formally comparable recovery actions, their residual-risk ordering, the preferred action when a formal recommendation is justified, and the declared support basis of that recommendation. The following subsections define these objects in the order in which they are used.

3.2 History-Dependent and Joint Operational State Formation

The recovery state describes what remains unresolved for the predecessor–successor operation at decision time t . Let

$$\mathcal{H}_{i,t} = (\mathcal{E}_{i,\tau})_{\tau \leq t} \quad (2)$$

denote the admissible operating history available for episode i up to the decision node. In application, $\mathcal{H}_{i,t}$ can include the observed progress of the predecessor flight, the freshness and quality of those observations, airport traffic and weather evolution, the successor schedule, and the remaining turnaround conditions. Section 4 specifies how this history is summarized empirically; Section 3 only requires that no information unavailable at time t enters the state.

The downstream recovery calculation depends on joint timing conditions rather than on expected delay alone. We therefore represent the unresolved state through aligned operating scenarios that retain the dependence among predecessor release, successor off-block execution, and taxi exposure. Prior aviation studies support the use of distributional delay outputs to represent unresolved timing uncertainty [Zoutendijk and Mitici, 2021, Dalmau et al., 2024]. The aligned joint scenario construction used here is a model specification of this study; Section 3.5 states the information-content requirement without attributing that joint structure to the forecasting literature.

Before the corresponding events become factual, the unresolved state contains three primitive quantities: the predecessor in-block time under the no-additional-action baseline, the successor off-block delay, and the successor excess taxi delay. For operating scenario s ,

$$\boldsymbol{\omega}_{i,t,s} = (T_{i,t,s}^{-,IB,0}, D_{i,t,s}^{+,OB}, D_{i,t,s}^{+,TX}), \quad \mathcal{S}_{i,t} = \{(\boldsymbol{\omega}_{i,t,s}, w_{i,t,s})\}_{s=1}^S, \quad (3)$$

where $w_{i,t,s}$ is the scenario weight. $T_{i,t,s}^{-,IB,0}$ indicates when the predecessor is expected to release the aircraft for the successor under the A00 baseline. $D_{i,t,s}^{+,OB}$ measures the successor’s off-block delay, and $D_{i,t,s}^{+,TX}$ measures the excess taxi delay beyond the operating reference. Together they describe the unresolved timing and ground-execution conditions that drive the subsequent recovery calculation.

The joint state follows the operating sequence from predecessor release to successor off-block execution and then taxi exposure:

$$p_t(t^{IB}, d^{OB}, d^{TX} \mid \mathcal{H}_{i,t}) = p_t(t^{IB} \mid \mathcal{H}_{i,t}) p_t(d^{OB} \mid t^{IB}, \mathcal{H}_{i,t}) p_t(d^{TX} \mid t^{IB}, d^{OB}, \mathcal{H}_{i,t}). \quad (4)$$

This factorization records an operational dependence rather than prescribing a particular statistical estimator: the successor cannot be evaluated independently of the predecessor release, and taxi exposure is interpreted conditional on the upstream in-block and off-block conditions of the same operating scenario.

Two quantities used repeatedly in the downstream recovery calculation are derived scenario by scenario:

$$R_{i,t,s}^{IB} = [T_{i,t,s}^{-,IB,0} - t]_+, \quad D_{i,t,s}^{+,TO} = D_{i,t,s}^{+,OB} + D_{i,t,s}^{+,TX}. \quad (5)$$

$R_{i,t,s}^{IB}$ is the remaining predecessor in-block exposure from the current decision time. $D_{i,t,s}^{+,TO}$ is the successor takeoff-delay exposure used in downstream consequence construction and is derived from the aligned off-block and taxi components rather than estimated independently. Turnaround shortfall, delay-threshold probabilities, and related operating indicators can then be calculated from $\mathcal{S}_{i,t}$ without replacing the scenario-level state itself.

As operations unfold, resolved uncertainty is replaced by observed facts rather than reconstructing the state from an isolated latest observation. Once predecessor in-block becomes factual, the observed value replaces its stochastic component and the successor quantities remain conditional on that fact. Once successor off-block becomes factual, the same update occurs again. This rolling contraction preserves aircraft continuity and the predecessor–successor dependence while reducing uncertainty as operational events are observed.

3.3 From Unresolved State to Operational Consequence

The unresolved operating state is not itself the recovery objective. A given predecessor–successor scenario can create different recovery consequences depending on the remaining turnaround margin, downstream schedule exposure, passenger exposure, and airport-resource conditions. For each operating scenario s , the state is therefore translated into a baseline operational-consequence profile

$$\omega_{i,t,s} \longrightarrow \mathbf{Q}_{i,t,s}^0.$$

The superscript 0 denotes the no-additional-action baseline from decision time t onward.

The consequence profile is organized into three airline-recovery channels and seven components. The flight-operation channel follows aircraft continuity, successor execution, and downstream propagation. The passenger-service channel separates passenger time, itinerary disruption, and service obligations. The recovery-resource channel records the additional operating burden associated with the disruption state. Define

$$\begin{aligned} \mathcal{K}_F &= \{F_{\text{cont}}, F_{\text{exec}}, F_{\text{prop}}\}, \\ \mathcal{K}_P &= \{P_{\text{time}}, P_{\text{itin}}, P_{\text{serv}}\}, \\ \mathcal{K}_R &= \{R_{\text{oper}}\}, \quad \mathcal{K} = \mathcal{K}_F \cup \mathcal{K}_P \cup \mathcal{K}_R. \end{aligned} \tag{6}$$

For $k \in \mathcal{K}$, let $Q_{i,t,s,k}^0$ denote the corresponding baseline operational consequence in its native unit. The seven components have different operational meanings. F_{cont} represents aircraft-continuity and turnaround exposure caused by delayed predecessor release. F_{exec} represents the successor’s off-block execution deviation. F_{prop} represents delay exposure carried into the downstream schedule. P_{time} records passenger time exposure; P_{itin} captures itinerary or connection disruption; P_{serv} records passenger-care and service obligations. R_{oper} represents additional operating-resource burden, including excess taxi and related ground-operation exposure. Section 4 specifies the empirical construction of these components.

The operational consequences are defined before monetary reporting. Constructed units provide a common measurement interface between heterogeneous airline consequences and a selected monetary system. For component k ,

$$C_{i,t,s,k}^{0,\text{CU}} = \psi_k(Q_{i,t,s,k}^0; \eta_k), \quad L_{i,t,s,k}^{0,m} = f_k^m(C_{i,t,s,k}^{0,\text{CU}}; \beta_k^m), \tag{7}$$

where $m \in \mathcal{M}$ indexes the monetary system. $Q_{i,t,s,k}^0$ and $C_{i,t,s,k}^{0,\text{CU}}$ do not depend on m ; changing the monetary system changes only the CU-to-money mapping f_k^m .

The scenario-level baseline consequence is aggregated only after the component meanings have been retained:

$$L_{i,t,s}^{0,m} = L_{i,t,s}^{F,m} + L_{i,t,s}^{P,m} + L_{i,t,s}^{R,m}, \tag{8}$$

with

$$\begin{aligned} L_{i,t,s}^{F,m} &= L_{i,t,s}^{F,\text{cont},m} + L_{i,t,s}^{F,\text{exec},m} + L_{i,t,s}^{F,\text{prop},m}, \\ L_{i,t,s}^{P,m} &= L_{i,t,s}^{P,\text{time},m} + L_{i,t,s}^{P,\text{itin},m} + L_{i,t,s}^{P,\text{serv},m}, \\ L_{i,t,s}^{R,m} &= L_{i,t,s}^{R,\text{oper},m}. \end{aligned}$$

This decomposition is retained because recovery actions can improve one operational component while worsening another. For example, protecting passenger itineraries can consume operating time or

resources, while an aircraft or ground intervention can reduce downstream propagation at the cost of additional recovery effort. The action comparison in Section 3.4 therefore acts on a common component structure rather than on a single undifferentiated delay measure.

3.4 Recovery Actions, State-Dependent Comparison, and Decision Output

Recovery action availability is determined by the current operating situation, not by the probabilistic state alone. A timing intervention, aircraft reassignment, passenger recovery action, or ground intervention may be meaningful only while its target remains identifiable, sufficient execution time remains, and the required operating conditions are satisfied. Recovery-action formation therefore uses current information directly while each available action is evaluated against the common state and consequence basis defined above.

Airline disruption recovery includes timing and capacity adjustments, passenger recovery, ground and aircraft interventions, crew recovery, and local-network changes [Hassan et al., 2021, Su et al., 2021, Ding et al., 2023, Davies et al., 2026]. The literature establishes these recovery families; the 23 templates below are the operational specification used in this study rather than a template set reproduced from any single source.

Table 2: Recovery-action library used to construct the episode-specific action set.

ID	Recovery action	Operational family	ID	Recovery action	Operational family
A00	No additional action	Baseline	A41	Gate/stand reassignment	Ground recovery
A11	Controlled connection hold	Timing/passenger	A42	Quick-turn ground-support reinforcement	Ground recovery
A13	Cruise-speed or flight-time adjustment	Flight execution	A43	Aircraft tow or stand reposition	Ground recovery
A21	Bounded departure retiming	Timing	A51	Same-fleet aircraft swap	Aircraft recovery
A22	Slot or ATFM timing coordination	Capacity	A52	Local rotation aircraft reassignment	Aircraft recovery
A23	Departure priority or sequence request	Capacity	A53	Standby aircraft activation	Aircraft recovery
A31	Passenger reaccommodation	Passenger recovery	A54	Cross-fleet equipment substitution	Aircraft recovery
A32	Expedited passenger transfer	Passenger recovery	A55	Ferry aircraft reposition	Aircraft recovery
A33	Passenger care/service activation	Passenger service	A61	Same-qualification crew swap	Crew recovery
A62	Reserve crew activation	Crew recovery	A63	Crew reposition or deadhead	Crew recovery
A64	Crew-duty repair or rescheduling	Crew recovery	A71	Successor-flight cancellation	Local network
A72	Downstream-leg cancellation or rotation truncation	Local network			

For each $a \in \mathcal{R}$, the methodology records a recovery-action specification

$$a \equiv (\iota_a, \theta_a, \mathcal{B}_a, \ell_a, \mathcal{K}_a^-, \mathcal{K}_a^+, \mathbf{G}_a, \Pi_a), \quad \mathbf{G}_a = \{G_{a,k}\}_{k \in \mathcal{K}}.$$

Here ι_a identifies the action and its operating target, θ_a contains episode-specific action parameters, $\mathcal{B}_a$ denotes the structural and operational preconditions, and ℓ_a is the minimum preparation or execution lead time. $\mathcal{K}_a^-$ and $\mathcal{K}_a^+$ identify consequence components that the action may reduce or increase. $\mathbf{G}_a$ specifies the component-level response, and Π_a records the declared basis supporting that response, such as published evidence, an operational rule, a scenario assumption, expert or engineering judgement, or a declared hybrid. Section 4 gives the empirical response forms and support categories.

The recovery-action library is independent of a particular episode. A new action a^* can be added as

$$\mathcal{R}^+ = \mathcal{R} \cup \{a^*\}, \quad (9)$$

provided that its target, required parameters, preconditions, execution time, affected consequence components, response model, and support basis are specified on the same terms as the existing actions.

The ability to extend the action library is therefore a consequence of using a common state and consequence basis rather than a separate cross-stage requirement.

Let $\chi_{i,t}(a) \in \{0, 1\}$ indicate whether action a is available at decision time t given the currently available flight, aircraft, timing, resource, and structural information. The episode-specific action set is

$$\mathcal{A}_{i,t} = \{a \in \mathcal{R} : \chi_{i,t}(a) = 1\}. \quad (10)$$

An action is excluded when a required target or parameter is unavailable, a known precondition is violated, or the remaining execution window is closed. An unresolved precondition can instead be retained for conditional assessment when the action is still meaningful for conditional evaluation.

The methodology distinguishes actions that can be numerically assessed from actions that can enter the formal recommendation. Let $\mathcal{A}_{i,t}^{\text{eval}} \subseteq \mathcal{A}_{i,t}$ contain actions for which the available structural information, execution opportunity, consequence coverage, and frozen response specification are sufficient to compute an action-conditioned consequence under the declared assumptions. Let $\mathcal{A}_{i,t}^{\text{cmp}} \subseteq \mathcal{A}_{i,t}^{\text{eval}}$ contain the actions admitted to the formal ranking. Membership in the latter set additionally requires the declared support basis to satisfy the comparison rule. Thus, a scenario- or expert-specified action may remain useful for conditional analysis without being presented as equally supported as an action whose response is grounded in stronger operational evidence.

For action a and consequence component k , the action modifies the same scenario-specific baseline consequence before monetary reporting:

$$C_{i,t,s,k}^{a,\text{CU}} = G_{a,k} \left(C_{i,t,s,k}^{0,\text{CU}}, \boldsymbol{\varepsilon}_{i,t}, \boldsymbol{\omega}_{i,t,s}; \boldsymbol{\theta}_a \right), \quad L_{i,t,s,k}^{a,m} = f_k^m \left(C_{i,t,s,k}^{a,\text{CU}}; \beta_k^m \right), \quad L_{i,t,s}^{a,m} = \sum_{k \in \mathcal{K}} L_{i,t,s,k}^{a,m}. \quad (11)$$

State dependence is preserved because each candidate action is evaluated on the same operating scenario $\boldsymbol{\omega}_{i,t,s}$ and the same baseline consequence profile. The action changes the consequence of that scenario; it does not construct a new local state for the purpose of scoring the action.

Residual recovery risk is summarized after the operating scenario, consequence scope, and comparison eligibility have been aligned:

$$J_{i,t}^{(\lambda,\alpha),m}(a) = (1 - \lambda) \mathbb{E}_w \left[L_{i,t}^{a,m} \right] + \lambda \text{CVaR}_{\alpha,w} \left(L_{i,t}^{a,m} \right), \quad 0 \leq \lambda \leq 1, \quad 0 < \alpha < 1. \quad (12)$$

The scenario weights $w_{i,t,s}$ are those retained from state formation. For monetary system m , the formal part of $\mathcal{D}_{i,t}$ orders $J_{i,t}^{(\lambda,\alpha),m}(a)$ over $a \in \mathcal{A}_{i,t}^{\text{cmp}}$. A00 provides the common no-additional-action baseline: $C_{i,t,s,k}^{A00,\text{CU}} = C_{i,t,s,k}^{0,\text{CU}}$ and $L_{i,t,s}^{A00,m} = L_{i,t,s}^{0,m}$. A00 means that the framework adds no recovery action from decision time t onward; it does not imply that the observed operation was historically free of intervention.

The decision output $\mathcal{D}_{i,t}$ is reported as an operational recovery result rather than a bare arg min. It retains the decision time and information cutoff, the current recovery-state and consequence profile, the actions available for the episode, the subset admitted to formal comparison, the residual-risk ranking, the preferred action and its improvement relative to A00, and the support status attached to the action response. Component-level consequences are retained so that the recommendation can be interpreted in terms of aircraft continuity, successor execution, downstream propagation, passenger exposure, and recovery-resource burden. Actions that are numerically assessable but not sufficiently supported for formal comparison are reported separately as conditional results.

3.5 Decision-Chain Consistency Conditions

The preceding subsections define how information, operating state, consequences, and recovery actions are used. The remaining requirement is that these objects continue to refer to the same rolling recovery problem as information moves through the calculation. The following conditions operationalize the

interface-consistency perspective introduced in Section 1 and connect it to the representation issue discussed in Section 2.

At a later decision node $t' > t$, newly available observations update $\mathcal{E}_{i,t'}$, the admissible operating history, the recovery state, and potentially the action set. The resulting $\mathcal{D}_{i,t'}$ is a new recovery comparison. Information observed after t does not alter the information boundary or state used to form $\mathcal{D}_{i,t}$.

1. *Information availability.* An operational observation can influence the recovery calculation at time t only after it is available to the decision process. If observation x becomes available at $t^{\text{avail}}(x)$, then

$$x \in \mathcal{E}_{i,t} \quad \Rightarrow \quad t^{\text{avail}}(x) \leq t. \quad (13)$$

This condition fixes the information boundary for state formation and action evaluation.

2. *Information-use continuity.* Information should retain the role for which it is needed downstream. Operating history used to characterize the current recovery situation remains represented through $\mathcal{H}_{i,t}$ and $\mathcal{S}_{i,t}$, whereas current execution, resource, and support information can remain directly available when $\mathcal{A}_{i,t}$ and $\mathcal{D}_{i,t}$ are formed. Information sharing therefore does not require all available information to be collapsed into one state vector.
3. *Scenario consistency.* Consequence construction and action evaluation must refer to the same operating scenario. For each s , the predecessor in-block, successor off-block, and taxi realization represented by $\omega_{i,t,s}$ remains the basis for $\mathbf{Q}_{i,t,s}^0$, $C_{i,t,s}^{0,\text{CU}}$, $C_{i,t,s}^{a,\text{CU}}$, and $L_{i,t,s}^{a,m}$. Downstream stages may transform this scenario, but they should not independently reconstruct a different local operating state.
4. *Common consequence basis.* Actions entering the same formal ranking must be evaluated over the same declared consequence components and aggregation rule. An action may affect only part of $\mathcal{K}$, but its formal comparison cannot omit consequence channels that remain part of the common recovery objective for other actions.
5. *Support transparency.* The declared basis supporting an operational input, consequence transformation, or action response remains explicit when that object is used downstream. Numerical transformation or optimization alone does not strengthen a scenario assumption, expert specification, or other support category.

These conditions also determine how much detail the operating-state representation must preserve. A point forecast can describe expected severity but cannot retain which predecessor-release, off-block, and taxi outcomes occur together. Separate marginal distributions preserve uncertainty in each quantity but do not, by themselves, retain the joint operating realization used by turnaround, propagation, passenger, and resource consequence calculations. The aligned scenario representation in Eq. (3) is therefore used because the downstream airline-recovery calculation requires these dependencies to remain identifiable. This is a requirement on decision-relevant information content, not a claim that one probabilistic estimator is universally necessary. An alternative representation would also be valid if it retained the same joint dependencies required by the downstream consequence and action calculations.

The same conditions apply when the action library is extended. A new recovery action can enter the decision process without redefining the upstream state only if it can be formed from decision-time information, expressed on the common consequence basis, evaluated against the same operating scenarios, and assigned an explicit support basis. The resulting methodology therefore retains the application-level ordering of information, operating state, consequence, intervention, and comparison while leaving the empirical forecasting and response models open to different implementations. Section 4 provides the implementation used in this study.

4 Empirical Specification of the Rolling Recovery Decision Chain

Section 3 defines the information, operating-state, consequence, and action-comparison objects of the rolling recovery model. This section specifies those objects in the principal empirical setting. Decision-time records first define the admissible operating information. The operating history then determines the unresolved predecessor–successor state. Each aligned operating scenario is translated into operational consequences, after which currently available recovery actions are evaluated on the same scenario and consequence basis. Estimation choices, literature-based mappings, action-response assumptions, and decision-policy parameters are fixed before final Test evaluation.

4.1 Decision-Time Operating Information and Empirical Cohort

The principal empirical dataset combines 2019 U.S. domestic operations from the BTS On-Time Performance records with NOAA Integrated Surface Database (ISD) observations, BTS DB1B and T-100 aggregate references, and static airport and time-zone information. We refer to this combination as the BTS–NOAA operating dataset. The sources have different roles in the recovery calculation rather than being treated as one undifferentiated feature table. Table 3 summarizes the information retained from each source.

Table 3: Decision-time and reference information used in the empirical recovery calculation.

Source	Principal information	Constructed recovery object	Use in the decision chain
BTS On-Time	Flight and aircraft identity; published schedule; gate and movement events; taxi durations	Predecessor–successor episode, event status, schedule timing, realized targets	Operating history, state estimation, factual contraction, and action targets
NOAA ISD	Observation time, temperature, dew point, wind, visibility, ceiling, supported supplementary weather	Decision-time airport-weather history with age and availability status	Dynamic operating environment for state estimation
BTS DB1B / T-100	Route passenger counts, segment passengers/seats, aircraft type where supported	Frozen passenger exposure V_{OD}^{pax} and segment/capacity references	Passenger consequences and applicable action context
Train-period BTS	Completed turnaround and taxi observations	Airport turnaround reference $T_a^{\text{turn,ref}}$ and taxi reference $T_a^{TX,\text{ref}}$	Aircraft-continuity, taxi-state, and resource consequences
BTS schedule chain	Tail identity and future scheduled departures	Downstream schedule exposure $E_{i,t}^{\text{down}}$ over a frozen 360-minute window	Delay-propagation consequence
Static references	Airport identifiers, coordinates, and time zones	Canonical airport identity and UTC timing basis	Source alignment and action-window timing
Passenger-recovery literature	Connection, rebooking, delay, and service relationships	Frozen mappings g_{itin} and g_{serv}	Itinerary and passenger-service consequence construction

A predecessor–successor episode is $i = (f_i^-, f_i^+)$. Two adjacent completed flight records form an empirical episode when they share the same non-missing aircraft identity, the predecessor destination equals the successor origin, and the realized gate gap satisfies

$$0 < T_i^{OB,+,act} - T_i^{IB,-,act} \leq 360 \text{ min.} \quad (14)$$

Pairs with cancelled or diverted flights, invalid temporal order, or a nonpositive scheduled turnaround window are excluded. Equation (14) is used only to identify the retrospective cohort and construct targets. The realized gate gap is not made available to an earlier rolling decision. The empirical results are therefore conditional on completed predecessor–successor chains satisfying this cohort definition rather than on the full population of cancellations, diversions, or very long-gap disruptions.

Rolling decision nodes are placed on a five-minute grid,

$$t_{i,n} = t_{i,0} + 5n, \quad (15)$$

with the information cutoff equal to the decision time at each node. Published schedule information is available from the start of the episode. A realized gate or movement event enters $\mathcal{E}_{i,t}$ only after that event has occurred and become available. This replay rule creates the decision-time information process required by Section 3.1 from a retrospective archive.

For NOAA observations, an airport-mapped observation with event time τ^{event} becomes available at

$$\tau^{\text{avail}} = \tau^{\text{event}} + 5 \text{ min}, \quad 0 \leq t - \tau^{\text{event}} \leq 60 \text{ min}. \quad (16)$$

At each decision node, the latest observation satisfying Eq. (16) is retained. Observation age, missingness, fallback status, and support status remain attached to the decoded weather variables. The operating history $\mathcal{H}_{i,t}$ can therefore distinguish a current observation from a stale, missing, or fallback value.

Airport turnaround and taxi references are estimated from the Train period when at least 50 valid observations are available; otherwise the declared Train-global fallback is used. Downstream exposure counts decision-visible same-aircraft scheduled departures within the frozen 360-minute window, and the passenger reference is frozen from the permitted DB1B reference period. These quantities are constructed once under the declared estimation rules and then reused downstream.

The time remaining to the successor’s scheduled off-block time is $c_{i,t} = T_i^{\text{OB},+,\text{sched}} - t$. For operating scenario s , the corresponding turnaround margin is

$$M_{i,t,s}^{\text{turn}} = c_{i,t} - R_{i,t,s}^{\text{IB}} - T_i^{\text{turn,ref}}, \quad (17)$$

where positive values denote remaining turnaround slack and negative values denote a shortfall relative to the successor’s scheduled off-block window.

The BTS–NOAA operating dataset does not contain continuous aircraft-surveillance trajectories or live crew, gate, standby-aircraft, and individual-slot states. Missing operational objects are not reconstructed from unrelated fields. Their absence instead limits which recovery actions can be treated as data-supported in a given episode; Section 4.4 makes this distinction explicit.

4.2 History-Aware Recovery-State Estimation

Section 3.2 defines $\mathcal{S}_{i,t}$ as a history-conditioned unresolved operating distribution represented by aligned operating scenarios $\omega_{i,t,s}$. The BTS–NOAA operating dataset estimates that state from the admissible operating history $\mathcal{H}_{i,t}$ using a principal recurrent specification and a non-recurrent reference specification. Both use the same decision-time information boundary, target definitions, chronological partitions, calibration procedure, and scenario-generation rule.

The principal state-aware estimator is a single-layer causal GRU. Its inputs are organized by their airline-operational role rather than treated as an undifferentiated feature vector: (i) observed predecessor and successor event status and elapsed-time information; (ii) current and recent airport-weather conditions; (iii) successor schedule and remaining-turnaround information; (iv) airport taxi and operating references; and (v) observation age, missingness, fallback, and support indicators. Static route, airport, aircraft, schedule, and reference information is retained with the recurrent summary. Continuous quantities are normalized using Train-period quantities only.

Let $x_{i,k}^{\text{hist}}$ denote the estimator input available at rolling node k . The estimator-specific recurrent state is

$$h_{i,k}^{\text{GRU}} = \text{GRU}_\eta \left(x_{i,k}^{\text{hist}}, h_{i,k-1}^{\text{GRU}} \right), \quad H = 32. \quad (18)$$

The notation $h_{i,k}^{\text{GRU}}$ is local to the estimator and is distinct from the admissible operating history $\mathcal{H}_{i,t}$ defined in Section 3.2. Recursive updates allow all admissible prior nodes to influence the current

estimate without treating later information as if it had been available earlier. A non-recurrent ARX–LightGBM distributional specification uses the same targets and information cutoff as the principal non-sequential reference. It provides a direct comparison for the value of recurrent operating history rather than a separate paper-level state definition.

The estimator produces the three primitive stochastic quantities fixed in Eq. (3): the predecessor in-block time under A00, the successor off-block delay, and the successor excess taxi delay. Empirically, the ordered conditional distributions are estimated as

$$\begin{aligned} \hat{p}\left(T_{i,t}^{-,IB,0} \mid \mathcal{H}_{i,t}\right), \\ \hat{p}\left(D_{i,t}^{+,OB} \mid T_{i,t}^{-,IB,0}, \mathcal{H}_{i,t}\right), \\ \hat{p}\left(D_{i,t}^{+,TX} \mid T_{i,t}^{-,IB,0}, D_{i,t}^{+,OB}, \mathcal{H}_{i,t}\right). \end{aligned} \quad (19)$$

The predecessor distribution is fitted with a discrete-hazard head, while the successor off-block and excess-taxi distributions use hurdle-quantile heads to accommodate a mass at zero and a continuous positive tail. The non-recurrent reference uses ARX–LightGBM heads with the same target semantics. Probability calibration is fitted on the Calibration period rather than the final Test period.

Aligned ancestral sampling implements the dependence in Eq. (4). For scenario s , the implementation first draws $T_{i,t,s}^{-,IB,0}$, then $D_{i,t,s}^{+,OB}$ conditional on that predecessor realization, and finally $D_{i,t,s}^{+,TX}$ conditional on both upstream realizations. The principal implementation uses $S = 1000$ scenarios, each retaining its scenario identifier and weight $w_{i,t,s}$. The operating scenario passed downstream is therefore exactly

$$\omega_{i,t,s} = \left(T_{i,t,s}^{-,IB,0}, D_{i,t,s}^{+,OB}, D_{i,t,s}^{+,TX}\right),$$

with the derived quantities

$$R_{i,t,s}^{IB} = [T_{i,t,s}^{-,IB,0} - t]_+, \quad D_{i,t,s}^{+,TO} = D_{i,t,s}^{+,OB} + D_{i,t,s}^{+,TX}. \quad (20)$$

$D_{i,t,s}^{+,TO}$ is a sample-aligned takeoff-delay exposure used in downstream consequence construction; it is not fitted as an independent stochastic head. When predecessor in-block, successor off-block, or another upstream event becomes factual, the observed value replaces the corresponding stochastic quantity and the remaining unresolved quantities continue conditionally. This is the empirical factual-contraction rule for $\mathcal{S}_{i,t}$.

The rolling interval, reporting horizon, and delay threshold are separate quantities. The principal decision chain updates every five minutes, with a 10-minute interval retained for sensitivity analysis. From the same terminal state distribution, operational reports query current and near-term event probabilities at

$$\mathcal{T} = \{0, 15, 60\} \text{ min}, \quad \mathcal{D} = \{15, 30, 60\} \text{ min} \quad (21)$$

for reporting horizons and delay-exceedance thresholds, respectively. The state distribution itself is not truncated at 60 minutes. Section 5 later evaluates predictions by the distinct decision-to-target lead time ℓ , including historical nodes up to 480 minutes before the target event; that evaluation lead time is not the reporting horizon $\tau \in \mathcal{T}$ used here.

4.3 Operational Consequence Construction and Monetary Reporting

Section 3.3 defines the baseline operational-consequence profile $\mathbf{Q}_{i,t,s}^0$ separately from the unresolved operating state. The BTS–NOAA operating dataset implements this mapping on each aligned scenario $\omega_{i,t,s}$ so that predecessor release, successor execution, taxi exposure, passenger exposure, and downstream schedule exposure remain attached to the same scenario identifier.

The flight-operation channel first distinguishes aircraft-continuity exposure, successor execution, and downstream propagation. With turnaround margin $M_{i,t,s}^{\text{turn}}$ from Eq. (17), the three native consequences

are

$$\begin{aligned} Q_{i,t,s}^{0,F,\text{cont}} &= [-M_{i,t,s}^{\text{turn}}]_+ = \left[R_{i,t,s}^{IB} + T_i^{\text{turn,ref}} - c_{i,t} \right]_+, \\ Q_{i,t,s}^{0,F,\text{exec}} &= [D_{i,t,s}^{+,OB}]_+, \\ Q_{i,t,s}^{0,F,\text{prop}} &= [D_{i,t,s}^{+,TO}]_+ E_{i,t}^{\text{down}}. \end{aligned} \quad (22)$$

The first quantity measures a loss of turnaround slack caused by delayed predecessor release, and the second records the successor's own off-block deviation. The third is a study-defined propagation proxy: it combines takeoff-delay exposure with the downstream same-aircraft schedule that remains exposed. Turnaround-aware recovery and delay-propagation studies motivate separating these mechanisms before valuation [Evler et al., 2022, Fleurquin et al., 2013]; the multiplicative form in Eq. (22) is the empirical operationalization used here rather than a formula reproduced from those studies.

The passenger-service channel distinguishes direct passenger time exposure from itinerary disruption and service obligations. With frozen route passenger exposure V_{OD}^{pax} ,

$$\begin{aligned} Q_{i,t,s}^{0,P,\text{time}} &= V_{OD}^{\text{pax}} [D_{i,t,s}^{+,TO}]_+, \\ Q_{i,t,s}^{0,P,\text{itin}} &= V_{OD}^{\text{pax}} g_{\text{itin}} \left(D_{i,t,s}^{+,TO}; \theta_{\text{itin}}^{\text{lit}} \right), \\ Q_{i,t,s}^{0,P,\text{serv}} &= V_{OD}^{\text{pax}} g_{\text{serv}} \left(D_{i,t,s}^{+,TO}; \theta_{\text{serv}}^{\text{lit}} \right). \end{aligned} \quad (23)$$

The first component is directly constructed from the aligned delay scenario and route-level passenger reference. The BTS–NOAA operating dataset does not observe individual passenger itineraries or realized passenger-care interventions. Passenger-recovery studies motivate explicit itinerary disruption and reaccommodation consequences, while delay-cost studies provide valuation and delay-band anchors [Bratu and Barnhart, 2006, Arikan et al., 2016, Ball et al., 2010, Cook and Tanner, 2015]. The itinerary and service mappings in Eq. (23) are therefore study-defined combinations of these declared anchors rather than equations taken from a single source. In the empirical implementation, $g_r(D) = \sum_h \theta_{r,h}^{\text{lit}} \phi_{r,h}(D)$ for $r \in \{\text{itin}, \text{serv}\}$, where $\phi_{r,h}$ is the declared delay-band or threshold basis. The basis, literature-anchored parameters, and any additional declared assumptions are frozen before final evaluation and recorded with their provenance.

The recovery-resource channel records additional operating burden associated with excess surface movement:

$$Q_{i,t,s}^{0,R,\text{oper}} = [D_{i,t,s}^{+,TX}]_+. \quad (24)$$

This quantity is a study-defined observable proxy for additional operating-resource exposure under the aligned scenario. Delay-cost references establish that additional operating delay has economic consequences [Cook and Tanner, 2015], but they do not define excess taxi minutes as this resource component; that proxy choice is specific to the present empirical construction.

These seven native quantities instantiate the component profile $Q_{i,t,s,k}^0$ for $k \in \mathcal{K}$ defined in Eq. (6). Each component is first mapped to the monetary-system-independent constructed-unit scale. The empirical normalization is

$$c_k^{\text{CU}} = \text{Median}_{\text{Train}} \left(Q_{i,t,s,k}^{0,\text{ref}} \mid Q_{i,t,s,k}^{0,\text{ref}} > 0 \right), \quad C_{i,t,s,k}^{0,\text{CU}} = \frac{Q_{i,t,s,k}^0}{c_k^{\text{CU}}}. \quad (25)$$

Here c_k^{CU} is a measurement-normalization reference, not a monetary coefficient. It places heterogeneous consequence components on a common constructed-unit interface without changing their operational meaning.

The empirical study then reports consequences in RMB using a frozen component-specific CU-to-money relationship,

$$L_{i,t,s,k}^{0,\text{RMB}} = \omega_k^{\text{RMB}} C_{i,t,s,k}^{0,\text{CU}}, \quad L_{i,t,s}^{0,\text{RMB}} = L_{i,t,s}^{F,\text{RMB}} + L_{i,t,s}^{P,\text{RMB}} + L_{i,t,s}^{R,\text{RMB}}. \quad (26)$$

Equation (25) is the empirical ψ_k in Eq. (7), whereas ω_k^{RMB} implements the RMB-specific f_k^{RMB} . The former defines how a native operational consequence is expressed in CU; the latter defines how one

CU of component k is reported in RMB. A change of monetary system therefore changes only the monetary mapping, not $\omega_{i,t,s}$, $\mathbf{Q}_{i,t,s}^0$, the CU normalization, or the action transformation.

Table 4 summarizes the seven components and their empirical basis.

Table 4: Operational consequence construction and empirical basis. References provide operational or valuation anchors; they do not necessarily define the exact empirical mapping used in this study.

Channel	Component	Operational consequence	Empirical basis	Operational / literature anchor
Flight	F_{cont}	Turnaround shortfall relative to the remaining scheduled-departure window	Aligned operating scenario + schedule timing + Train turnaround reference	Turnaround mechanism: [Evler et al., 2022]
Flight	F_{exec}	Positive successor off-block deviation	Aligned operating scenario	Delay-cost rationale: [Ball et al., 2010, Cook and Tanner, 2015]
Flight	F_{prop}	Study-defined takeoff-delay/downstream-schedule propagation proxy	Aligned operating scenario + frozen schedule chain	Propagation mechanism: [Fleurquin et al., 2013, Evler et al., 2022]
Passenger	P_{time}	Passenger-minutes of delay exposure	Aligned operating scenario + route passenger reference	Passenger-delay rationale: [Ball et al., 2010]
Passenger	P_{itin}	Study-defined itinerary-disruption mapping	Passenger reference + frozen declared mapping	Recovery mechanism: [Bratu and Barnhart, 2006, Arian et al., 2016]
Passenger	P_{serv}	Study-defined passenger-care/service mapping	Passenger reference + frozen declared mapping	Delay-cost/service anchor: [Ball et al., 2010, Cook and Tanner, 2015]
Resource	R_{oper}	Study-defined excess-taxi resource proxy	Aligned excess-taxi scenario	Delay-cost rationale: [Cook and Tanner, 2015]

The component decomposition is preserved when recovery actions are evaluated. A reduction in one channel and an induced consequence in another therefore remain visible until the common residual-risk aggregation, rather than being collapsed before the action comparison.

4.4 Recovery-Action Instantiation and State-Dependent Comparison

The recovery-action library $\mathcal{R}$ contains the 23 templates defined in Section 3.4, including the no-additional-action baseline $A00$. The BTS-NOAA operating dataset does not define the library; it determines which action targets, preconditions, execution windows, and support information are observable at a particular decision node. The empirical calculation therefore evaluates the availability indicator $\chi_{i,t}(a)$ in Eq. (10) using the current flight, aircraft, timing, resource, and structural information in $\mathcal{E}_{i,t}$.

Table 5 distinguishes information that is directly observed from information that remains partial or unavailable in the public records. The table concerns the information needed to instantiate an action, not the effectiveness of that action, which is handled separately through Π_a and the frozen response specification.

Table 5: Information support for recovery-action families in the principal empirical dataset.

Action family	Information directly available	Information not directly observed	Empirical treatment
Timing / connection hold	Schedule, flight identity, current event status, passenger exposure reference	Individual connection state where required	Timing is instantiated directly; passenger-specific preconditions remain conditional when not observed
Flight execution / capacity	Schedule, event timing, airport context	Live ATFM slot, sequence priority, controller authorization	Evaluated only when the required capacity precondition has declared external or conditional support
Passenger recovery / service	Aggregate route passenger exposure and delay state	Individual itineraries, booking status, realized reaccommodation options	Consequences use aggregate frozen mappings; individual-itinerary actions remain conditional where required
Ground recovery	Airport identity and timing context	Live gate, stand, tow, and ground-resource availability	Not treated as data-supported from the public records alone; requires declared conditional or external support
Aircraft recovery	Tail identity, rotation context, aircraft type where supported	Spare/standby aircraft, cross-fleet compatibility, real-time fleet availability	Structural target can be observed, but resource feasibility requires additional support
Crew recovery	Flight timing and aircraft context	Live crew duty, qualification, reserve status, reposition feasibility	Not treated as data-supported from the public records alone; retained only under declared conditional or external support
Local-network adjustment	Schedule chain and downstream flight identity	Operational cancellation authority and action-specific network response	Structural target is observed; response and formal comparison depend on the declared support basis

For an action retained in $\mathcal{A}_{i,t}$, the empirical calculation resolves the episode-specific target and those parameters θ_a that are supported by the available records. Observable preconditions $\mathcal{B}_a$ are checked directly. Resource-specific preconditions that are not observed are not inferred; they restrict the action to the corresponding conditional or externally supported lane. The affected consequence components $\mathcal{K}_a^-$ and $\mathcal{K}_a^+$ and the declared support basis Π_a remain attached to the action. The remaining execution opportunity is then evaluated on the aligned operating scenarios. For action a ,

$$O_{i,t,s}(a) = \mathbb{I}\left[T_{i,t,s}^{\text{deadline}(a)} > t + \ell_a\right], \quad \pi_{i,t}^{\text{opp}}(a) = \sum_{s=1}^S w_{i,t,s} O_{i,t,s}(a). \quad (27)$$

The operational meaning of $T_{i,t,s}^{\text{deadline}(a)}$ depends on the action family: a turnaround or ground action must remain executable before the relevant ground-process deadline; a departure-timing action must remain open before successor departure or takeoff; and a passenger or downstream action is tied to the corresponding connection, service, or downstream schedule opportunity. An action with no remaining modeled opportunity is excluded from current numerical evaluation and formal comparison.

For the 22 non-null templates, action-response uncertainty is represented by a frozen Bernoulli–Beta specification,

$$Z_{i,t,s}(a) \sim \text{Bernoulli}(p_a), \quad B_{i,t,s}(a) \mid Z_{i,t,s}(a) = 1 \sim \text{Beta}(\alpha_a, \beta_a), \quad \rho_{i,t,s}(a) = Z_{i,t,s}(a)B_{i,t,s}(a). \quad (28)$$

The public operational records do not identify action-specific causal treatment effects. The literature cited for recovery-action families establishes that such intervention classes are operationally relevant; it does not identify the numerical response assigned to each of the 22 non-null templates in this study. The parameters (p_a, α_a, β_a) are therefore externally specified from published evidence, operational rules, scenario assumptions, expert or engineering judgement, or a declared combination of these sources. Their values and provenance are retained in Π_a and frozen before final evaluation. Equation (28) supplies the uncertainty used by the empirical action-response function; it does not by itself make an action formally comparable.

Recovery actions modify the same CU consequence attached to the same operating scenario. For an open non-null action, mitigation coefficient $m_{a,k}$, induced-consequence score $d_{a,k}$, response intensity $\rho_{i,t,s}(a)$, and frozen induced-consequence conversion γ give

$$C_{i,t,s,k}^{a,\text{CU}} = \max\left\{0, C_{i,t,s,k}^{0,\text{CU}}(1 - m_{a,k}\rho_{i,t,s}(a))\right\} + \gamma d_{a,k}. \quad (29)$$

Equation (29) is the empirical form of $G_{a,k}$ in Eq. (11). The action does not generate a new operating state; it changes the consequence of $\omega_{i,t,s}$ under the declared response specification. Under RMB reporting,

$$L_{i,t,s,k}^{a,\text{RMB}} = f_k^{\text{RMB}}\left(C_{i,t,s,k}^{a,\text{CU}}; \beta_k^{\text{RMB}}\right), \quad L_{i,t,s}^{a,\text{RMB}} = \sum_{k \in \mathcal{K}} L_{i,t,s,k}^{a,\text{RMB}}. \quad (30)$$

For A00, $C_{i,t,s,k}^{A00,\text{CU}} = C_{i,t,s,k}^{0,\text{CU}}$, so the no-additional-action baseline is an exact samplewise identity before monetary reporting.

Comparison follows the hierarchy fixed in Section 3.4. $\mathcal{A}_{i,t}^{\text{eval}}$ contains available actions for which structural information, execution opportunity, consequence coverage, and a frozen response specification are sufficient to calculate $C_{i,t,s,k}^{a,\text{CU}}$. $\mathcal{A}_{i,t}^{\text{cmp}} \subseteq \mathcal{A}_{i,t}^{\text{eval}}$ contains the actions admitted to the formal ranking after the declared support rule is applied. Formally compared actions use the same decision-time information cutoff, scenario identifiers s , weights $w_{i,t,s}$, baseline consequence $C_{i,t,s,k}^{0,\text{CU}}$, seven-component consequence scope, and RMB reporting interface. Numerically evaluable actions that do not satisfy the formal support rule remain available for conditional analysis but do not fill positions in the formal ranking.

The principal residual-risk policy fixes $\lambda = 0.25$ and $\alpha = 0.90$. Equation (12) is therefore implemented as

$$J_{i,t}^{(0.25,0.90),\text{RMB}}(a) = 0.75 \mathbb{E}_w \left[L_{i,t}^{a,\text{RMB}} \right] + 0.25 \text{CVaR}_{0.90,w} \left(L_{i,t}^{a,\text{RMB}} \right). \quad (31)$$

The formal part of $\mathcal{D}_{i,t}$ orders $a \in \mathcal{A}_{i,t}^{\text{cmp}}$ under Eq. (31). The reported recovery result retains the decision time and information cutoff, the current state and consequence exposure, the available and formally comparable actions, the residual-risk ranking, the preferred action and its improvement relative to A00, component-level consequences, and the support basis attached to the recommended action. Conditional results are reported separately when a numerical action evaluation is possible but formal recommendation is not supported.

4.5 Estimation Protocol and Rolling Decision Operation

The BTS–NOAA operating dataset is partitioned chronologically by successor service date. Train covers January–June 2019, Calibration July, Development August–September, and Test October–December. The partitions have different roles. Train fits the state estimators and Train-derived operating references and CU normalization quantities. Calibration is used only for probability calibration. Development fixes estimator architecture and tuning choices, literature-parameterized consequence mappings, the action-response registry, support rules, and residual-risk policy. Test is reserved for final out-of-sample evaluation.

The frozen quantities in this section come from three distinct sources. *Data-estimated quantities* include Train-period turnaround and taxi references, CU normalization scales, fitted estimator parameters, and Calibration-period probability adjustments. *Development-selected quantities* include the principal rolling interval, GRU dimension, aligned-scenario count, reporting settings, and residual-risk parameters. *Externally specified quantities* include literature-based passenger mappings, the action-response parameters and their provenance, and the RMB monetary-reporting coefficients. This distinction prevents an externally assumed operational response from being interpreted as an effect estimated from the Test data.

Table 6 summarizes the principal empirical specification.

Table 6: Principal empirical specification of the rolling recovery calculation.

Item	Principal specification	Frozen / estimated from
Rolling interval	5 minutes; 10 minutes for sensitivity analysis	Development
Principal state estimator	One-layer causal GRU	Train + Development
Non-recurrent reference	ARX-LightGBM distributional estimator	Train + Development
Primitive stochastic outputs	$T^{-,IB,0}, D^{+,OB}, D^{+,TX}$	Train targets
Derived takeoff exposure	$D^{+,TO} = D^{+,OB} + D^{+,TX}$ samplewise	Fixed definition
Distribution heads	In-block discrete hazard; successor hurdle-quantile heads	Development
Aligned operating scenarios	$S = 1000$	Development
Reporting horizons	$\{0, 15, 60\}$ minutes	Development
Delay thresholds	$\{15, 30, 60\}$ minutes	Development
Consequence scope	Three channels / seven components	Model definition
Passenger itinerary/service mappings	Passenger exposure + frozen literature mappings	External, pre-Test
CU normalization	Positive Train-period component references	Train
Monetary reporting	CU $\rightarrow$ RMB component mapping	External / Development
Recovery library	23 templates including A00	Model definition
Action-response model	Bernoulli-Beta for 22 non-null templates	External, pre-Test
Comparison hierarchy	$\mathcal{A}_{i,t} \rightarrow \mathcal{A}_{i,t}^{\text{eval}} \rightarrow \mathcal{A}_{i,t}^{\text{cmp}}$	Fixed rule
Residual-risk parameters	$\lambda = 0.25, \alpha = 0.90$	Development

At each five-minute node, newly admissible information updates $\mathcal{E}_{i,t}$ and the admissible operating history $\mathcal{H}_{i,t}$. The recovery-state estimator then updates $\mathcal{S}_{i,t}$, with newly factual events replacing resolved stochastic quantities. The same scenario identifiers are carried into $\mathbf{Q}_{i,t,s}^0$ and $C_{i,t,s,k}^{0,\text{CU}}$. Current information can change action availability and execution opportunity, after which surviving actions are reevaluated on the updated common scenario and consequence basis. Train-, Calibration-, Development-, and externally specified quantities remain fixed during Test. Rolling adaptation therefore occurs through newly available operating information, factual state contraction, and changing recovery opportunity rather than through test-time refitting of the model or retrospective revision of earlier decisions.

5 Experimental Design and Evaluation

The experiments examine four operational questions that follow from the proposed recovery decision chain. First, does rolling recovery require both direct reuse of current operational information and retention of admissible history? Second, must the operating state and its consequences preserve uncertainty, dependence, and mechanism-specific structure for recovery comparison? Third, as new information arrives, should recovery recommendations be refreshed and should the state propagated downstream remain synchronized with the current decision time? Finally, does the complete recovery framework provide adequate predictive performance, operationally supportable recommendations, cross-data robustness, and computational performance? The first three experiments isolate these methodological requirements. The fourth evaluates the complete recovery framework.

5.1 Experimental Design

Experiments 1–3 use controlled comparisons in which one element of the decision chain is changed while the remaining model components are kept unchanged. This separation is important because the experiments are intended to identify the role of information retention, representation, and temporal updating rather than differences in action spaces or objective functions. Experiment 4 then evaluates the complete recovery framework without introducing a further methodological ablation.

The primary empirical environment is Data2, using the chronological data partition described in Section 4.5. Statistical comparisons are paired at the episode level so that multiple rolling decision nodes from the same predecessor–successor episode are not treated as independent observations. Confidence intervals are obtained by episode-level bootstrap resampling.

Some controlled comparisons require a common evaluation of actions selected under different experimental conditions. After the realized episode outcome becomes available, an action a is evaluated as

$$J_i^{\text{post}}(a) = \mathcal{J}^{\text{eval}}(a; Y_i^{\text{obs}}), \quad (32)$$

where Y_i^{obs} denotes realized evaluation information. This quantity is used only for ex-post comparison. It is a model-implied evaluation under the common action-response specification, rather than an observed causal treatment effect.

Table 7: Experimental questions and principal comparisons.

Experiment	Question	Principal comparison	Main evidence
Exp1	Which information roles must be retained?	Direct reuse vs no direct reuse; current state vs history-conditioned state	Prediction and decision output
Exp2	Which representation must be retained across stages?	Point–Marginal–Joint; Scalar–3-channel–7-component	Proper scores and decision change
Exp3	How should the chain operate as information evolves?	One-shot vs rolling; synchronized vs lagged state	Executability and ex-post comparison
Exp4	Is the complete recovery framework empirically and operationally adequate?	Prediction, recommendation validity, cross-data comparison, and runtime	Observed and operational evidence

5.2 Value of Cross-Stage Information Retention

Experiment 1 tests the two information roles formalized in Sections 3.1 and 3.5. Historical information can be retained through the history-conditioned operating state, whereas current information can remain directly relevant when recovery actions are formed and compared. The two roles are examined separately.

5.2.1 Direct Reuse of Current Operational Information

The first comparison asks whether the history-conditioned operating state is sufficient for recovery comparison, or whether current operational information must also remain directly available when feasible recovery alternatives are formed and compared. The Full condition follows the formal information-sharing structure. The restricted condition retains only the current facts required to define the available recovery alternatives and their operational feasibility.

Let $P_{\text{req}}(\mathcal{E}_{i,t})$ denote this restricted current-information set. The two decision constructions are

$$\begin{aligned}\mathcal{A}_{i,t}^F &= F_A(\mathcal{E}_{i,t}, \mathcal{R}), & \mathcal{D}_{i,t}^F &= F_D(\mathcal{E}_{i,t}, \mathcal{S}_{i,t}, \mathcal{A}_{i,t}^F), \\ \mathcal{A}_{i,t}^N &= F_A(P_{\text{req}}(\mathcal{E}_{i,t}), \mathcal{R}), & \mathcal{D}_{i,t}^N &= F_D(P_{\text{req}}(\mathcal{E}_{i,t}), \mathcal{S}_{i,t}, \mathcal{A}_{i,t}^N).\end{aligned}\tag{33}$$

Both conditions retain the same state construction, consequence model, recovery library, and comparison criterion. The comparison therefore isolates the operational role of direct information reuse without changing the sequence of state formation, consequence assessment, and recovery-action comparison.

The principal endpoint is the change in the Top-1 recovery recommendation. Equation (32) provides a common ex-post comparison when the two conditions select different actions.

5.2.2 History-Conditioned Operating State

The second comparison asks whether the latest admissible snapshot is sufficient to represent the unresolved airline operating condition. The Current condition uses only the information at the current decision node, whereas the History condition uses the admissible information sequence available up to that node:

$$\mathbf{h}_{i,t}^{\text{cur}} = H_\eta(\mathcal{E}_{i,t}), \quad \mathbf{h}_{i,t}^{\text{hist}} = H_\eta(\mathcal{E}_{i,\leq t}).\tag{34}$$

The predictive model form and the downstream consequence and action-comparison procedures are otherwise held fixed. A finite-history comparator can be added as an intermediate condition if required.

CRPS is used to evaluate the primitive stochastic outputs, with a delay-event Brier score as a secondary probabilistic measure. Downstream recommendation disagreement is then used to determine whether differences in state formation propagate into recovery decisions.

5.3 Representation of the Operating State and Recovery Consequences

Experiment 2 keeps the information available at the decision time unchanged and varies only its representation. The first comparison examines whether uncertainty and cross-variable dependence must be retained in the operating state. The second examines whether consequence mechanisms must remain identifiable rather than being aggregated before action comparison.

5.3.1 Uncertainty and Dependence Representation

Let the formal operating-state scenario be

$$\mathbf{z}_{i,t,s} = (R_{i,t,s}^{IB}, \Delta_{i,t,s}^{OB}, T_{i,t,s}^{TX}).$$

The Joint condition retains the aligned scenarios. The Marginal condition preserves the marginal distributions of the primitive quantities but removes their original cross-variable alignment:

$$\mathbf{z}_{i,t,s}^M = (\Pi_1[R^{IB}]_{i,t,s}, \Pi_2[\Delta^{OB}]_{i,t,s}, \Pi_3[T^{TX}]_{i,t,s}), \quad \hat{F}_j^M = \hat{F}_j^J.\tag{35}$$

Thus, any difference between the Marginal and Joint conditions is attributable to the loss of dependence rather than to different primitive marginal distributions.

The Point condition removes distributional uncertainty while retaining one coherent operating realization. It is represented by a central joint scenario rather than by independently combining marginal summaries.

CRPS evaluates the marginal predictive distributions. Dependence-sensitive multivariate quality is evaluated by the Variogram Score,

$$\text{VS}_p(F, \mathbf{y}) = \sum_{j < k} h_{jk} (|y_j - y_k|^p - \mathbb{E}_F |X_j - X_k|^p)^2, \quad p = 0.5.\tag{36}$$

The decision-level endpoint is Top-1 disagreement with the Joint representation. This measures whether simplifying the operating-state representation changes the recovery recommendation; it does not treat the Joint recommendation as an observed ground-truth action.

5.3.2 Representation of Recovery Consequences

The second representation comparison asks whether total consequence is sufficient for recovery comparison, or whether the mechanisms that generate that consequence must remain identifiable. Starting from the seven-component consequence vector, three levels of representation are considered:

$$P_r(\mathbf{C}) = \begin{cases} \mathbf{C}, & r = 7, \\ \left(\sum_{k \in \mathcal{K}_F} C_k, \sum_{k \in \mathcal{K}_P} C_k, \sum_{k \in \mathcal{K}_R} C_k \right), & r = 3, \\ \sum_{k \in \mathcal{K}} C_k, & r = 1. \end{cases} \quad (37)$$

These correspond to Seven-Component, Three-Channel, and Scalar representations. The same upstream operating state and recovery alternatives are used in all three conditions.

The principal endpoint is the proportion of decision opportunities for which the selected recovery action differs from that obtained under the seven-component representation. Action-family composition is used to show whether aggregation shifts the recovery response across timing, passenger, ground, aircraft, crew, and network-level interventions. A matched-case comparison further examines episodes with similar aggregate disruption but different consequence composition, providing a direct test of whether mechanism-specific information affects the intervention choice.

5.4 Rolling Recommendation Refresh and State Synchronization

Experiment 3 keeps the information and representation contracts unchanged and varies only the temporal use of the decision chain. The objective is not to establish rolling recovery as a new concept, but to test whether new information must be propagated through the proposed chain and whether the operating state used downstream must remain aligned with the current decision time.

5.4.1 Recommendation Refresh

For episode i , the first node at which a genuine recovery choice can be made is defined as

$$t_i^0 = \min \left\{ t : |\mathcal{A}_{i,t}^{\text{cmp}}| \geq 2, \mathcal{A}_{i,t}^{\text{cmp}} \setminus \{A00\} \neq \emptyset \right\}. \quad (38)$$

The One-Shot condition retains the recommendation formed at t_i^0 . The Rolling condition updates the recommendation as new admissible information enters the subsequent decision nodes. The model, action space, and comparison criterion are unchanged.

The main operational endpoint is the proportion of initially valid recovery recommendations that remain executable as the recommendation ages. When both One-Shot and Rolling recommendations remain comparable on a common basis, Eq. (32) is used as a secondary quantitative comparison.

5.4.2 State Synchronization

The second temporal comparison asks whether a refreshed decision must use an operating state aligned with the current decision time. Current information and the current action set are held fixed, while only the state vintage changes:

$$\mathcal{D}_{i,t}^{(\delta)} = F_D(\mathcal{E}_{i,t}, \mathcal{S}_{i,t-\delta}, \mathcal{A}_{i,t}), \quad \delta \in \{0, 5, 10\} \text{ minutes}. \quad (39)$$

The synchronized condition is $\delta = 0$. The comparison therefore isolates the effect of using an outdated rather than time-aligned operating state for the current recovery decision.

The principal endpoints are the selected-action difference and the ex-post comparison in Eq. (32). A small number of operational strata, including turnaround flexibility and the type of information update, are used to examine where state synchronization is most consequential.

5.5 Overall Empirical and Operational Adequacy

Experiments 1–3 test why the decision chain is organized as proposed. Experiment 4 instead evaluates the complete recovery framework. Four dimensions are considered: predictive adequacy, operational admissibility, portability across evidence environments, and computational adequacy. These checks are intended to establish whether the framework can support rolling recovery analysis; they do not identify the causal effect of unexecuted recovery actions.

5.5.1 Predictive Adequacy across the Operational Lead-Time Window

Prediction is evaluated by the time remaining from a historical decision node to the target operational event:

$$\ell_{i,t} = T_i^{\text{target}} - t, \quad \ell \in \{0, 30, 60, 120, 180, 240, 300, 360, 420, 480\} \text{ minutes.} \quad (40)$$

The evaluation lead time ℓ is distinct from the forecast horizon τ used by the state estimator.

The principal predictive target is the successor flight’s total takeoff delay. The complete history-conditioned model is compared with a historical reference and strong non-sequential tabular baselines. The methods share the same target definition, information cutoff, chronological evaluation period, and observed outcome.

MAE in minutes and CRPS provide the principal point and distributional measures, respectively. The aim is to establish predictive adequacy across the operational decision window rather than to turn the paper into a forecasting-model benchmark.

5.5.2 Operational Admissibility of Recovery Recommendations

The second adequacy test asks whether a formal recommendation is supportable at the time it is issued. Let $\mathcal{N}^{\text{elig}}$ denote the eligible rolling decision nodes. Formal recovery recommendation availability is

$$R_{\text{formal}} = \frac{\#\{(i, t) \in \mathcal{N}^{\text{elig}} : |\mathcal{A}_{i,t}^{\text{cmp}}| \geq 2, \mathcal{A}_{i,t}^{\text{cmp}} \setminus \{A00\} \neq \emptyset\}}{|\mathcal{N}^{\text{elig}}|}. \quad (41)$$

The denominator therefore represents decision opportunities rather than only cases in which a recommendation was already available.

Formal recommendations are then evaluated separately for execution feasibility, structural feasibility, factual consistency, evidential support, and decision-time information consistency. These quantities remain separate because they describe different conditions of admissibility rather than components of a single performance index.

5.5.3 Cross-Data Robustness and Support Portability

Data2 is the primary empirical environment, while Data1 provides a second operational evidence environment with weaker observability. The portability analysis is restricted to targets and lead times with common operational meaning across the two datasets.

Predictive portability is assessed primarily through the relative performance of the complete model and the same non-sequential reference within each data environment. This avoids interpreting raw Data1–Data2 error differences as a portability effect when the two environments differ in observability. Decision-support portability is assessed by the extent to which formal decision support is preserved, reduced, or unavailable under the weaker evidence environment.

5.5.4 Computational Adequacy

The final experiment asks whether the complete recovery framework can produce a recovery recommendation, or explicitly withhold one, within the five-minute rolling cycle. End-to-end latency is therefore evaluated for the complete decision chain rather than for an isolated prediction module.

The principal summaries are median, p95, and p99 latency. The five-minute update interval provides the operational reference for judging whether the computation is compatible with the intended rolling use. Detailed stage-level timing is reserved for supporting analysis rather than treated as a separate scientific contribution.

6 Discussion

7 Conclusion

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